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RESEARCH ARTICLE

CNN-LiDAR-SLAM: Multimodal Fusion With Object Detection for Indoor Localization

SAQI HUSSAIN KALAN¹, BOON GIIN LEE², (Senior Member, IEEE),
AND WAN-YOUNG CHUNG¹, (Senior Member, IEEE)

¹Department of Artificial Intelligence Convergence, Pukyong National University, Nam-gu, Busan 48513, Republic of Korea

²School of Computer Science, University of Nottingham Ningbo China, Ningbo 315100, China

Corresponding author: Wan-Young Chung (wychung@pkn.ac.kr)

ABSTRACT Simultaneous localization and mapping (SLAM) in indoor environments remains a critical challenge, especially in GPS-denied areas where real-time Performance, low cost, and robustness are required. Traditional 3D LiDAR-based solutions offer accuracy but are expensive and computationally demanding. This paper presents CNN-LIDAR-SLAM, a lightweight, handheld approach that integrates 2D LiDAR, IMU, and a CNN-RNN-based object detection framework. By combining inertial Odometry with learned semantic landmarks, and optimizing with an Iterated Extended Kalman Filter (IEKF), our approach delivers robust and precise localization without relying on loop closure. The proposed system achieves consistent Performance in cluttered, dynamic environments and operates in real-time on modest hardware. Our results validate the system's effectiveness as a cost-efficient, accurate, and scalable solution for autonomous navigation and indoor mapping.

INDEX TERMS SLAM, odometry, mapping, object detection, CNN, indoor localization, sensor fusion, autonomous navigation.

I. INTRODUCTION

Indoor localization has emerged as an essential technology in diverse domains such as robotics, industrial automation, and rescue missions like [1]. Both autonomous and handheld devices require accurate mapping and navigation in intricate indoor settings to function effectively in real-time situations. However, achieving precise localization and mapping in dynamic and unstructured environments is challenging due to obstacles, fluctuating conditions, and sensor constraints [2]. Simultaneous localization and mapping (SLAM) is frequently applied in unfamiliar environments, using LiDAR for real-time perception, localization, and mapping of mobile robots or handheld gadgets. Among various SLAM methodologies, LiDAR-based SLAM [3] is lauded for its high precision in structured environments. In contrast, visual-SLAM, although more cost-effective, faces difficulties with reduced accuracy and robustness in dynamic scenarios [4].

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Perception is critical in ensuring safe navigation in emergency operations in various indoor environments [5]. LiDAR technology, which calculates the flight time of light pulses to measure distances, provides the high-resolution data necessary for accurate object identification and distance estimation [3]. Despite its advantages, including precise measurement capabilities, 3D LiDAR is limited by its high cost, restricting its widespread adoption [6]. In contrast, 2D LiDAR offers a more affordable option while supporting multidimensional object detection [7]. Recent studies have focused on improving the reliability and accuracy of localization methods by integrating 2D LiDAR data with additional sensors, such as inertial measurement units (IMUs) [4].

The broad application of 2D LiDAR is attributed to its 360-degree horizontal field of view, accurate distance measurement, and immunity to ambient lighting and optical textures, which allows for effective perception even in dark environments. However, LiDAR-based techniques encounter substantial difficulties in settings such as long corridors, tunnels, and open roads, where their Performance is degraded [3]. This degradation often leads to inaccurate

state estimation and overlapping maps in SLAM methods. In large-scale or complex environments, single sensors may be insufficient to ensure high-precision SLAM due to the high demands on data processing, real-time Performance, and system stability. Studies have shown that multisensor fusion significantly improves the accuracy and robustness of SLAM systems, particularly when integrating LiDAR and IMU data [8], [9]. However, multisensor settings' synchronization, calibration, and learning stages continue to be intricate due to differences in frequency, accuracy, and coordinate frameworks between 3D LiDAR and IMU sensors [10].

Earlier studies have explored 2D LiDAR-camera fusion for cost-effective localization and obstacle detection [11]. However, the real-time efficacy of this integration in environments and the potential advances through sophisticated sensor fusion remain underexploited. Recent studies highlight the limitations of LiDAR SLAM methods in indoor settings, where data volume and complexity from multiple sensors demand significant computational power. Challenges like sensor drift and continuous inaccuracies obstruct long-term system functionality [3], [12].

Despite the remarkable progress in LiDAR- and vision-based SLAM, a distinct research gap persists between high-precision 2D SLAM systems and lightweight, low-cost indoor solutions. Most existing approaches either depend on computationally intensive 3D LiDAR sensors or rely on vision-based methods that degrade under poor lighting or dynamic conditions. Furthermore, limited research has explored real-time fusion of 2D LiDAR and IMU data with deep neural object detection to achieve semantic awareness on embedded hardware. This gap highlights the need for a unified, resource-efficient SLAM framework capable of maintaining high localization accuracy and robustness in cluttered, GPS-denied indoor environments without relying on loop closure or GPU acceleration.

This paper introduces a novel CNN-LiDAR-SLAM system that significantly enhances indoor localization accuracy by integrating 2D LiDAR, IMU sensors, and a CNN-based object detection algorithm. Unlike conventional SLAM systems, our approach leverages real-time CNN-driven object detection to improve mapping precision and localization accuracy. By optimizing sensor fusion through iterated extended Kalman filters (IEKF) and FAST-LIO2 [8], the system reduces Absolute Trajectory Error (ATE) by 23%, a considerable improvement over state-of-the-art techniques. This integration system is designed to handle dynamic and cluttered indoor environments, which are often challenging for traditional methods. We make three claims: (i) A lightweight semantic SLAM framework combining 2D LiDAR, IMU, and compact CNN-based object detection, designed as a cost-effective alternative to traditional 3D LiDAR-based systems. This tightly coupled approach enables accurate indoor localization while reducing hardware complexity and power requirements. (ii) An adapted IEKF-based sensor fusion backend incorporating semantic object landmarks. Unlike conventional methods that rely solely on geometric

features, our system integrates semantic cues into the state estimation pipeline, improving localization robustness in feature-sparse or dynamic scenes without requiring full graph optimization. (iii) A real-time implementation on embedded hardware, demonstrating that semantic SLAM with object-aware mapping can be achieved using only 2D LiDAR and IMU on a Raspberry Pi 4 B. Our system runs at 15 FPS with under 40% CPU usage and no GPU acceleration, validating its deployment potential in mobile or wearable indoor robotics. In contrast to prior semantic SLAM frameworks such as DynaSLAM, ORB-SLAM3 with semantics, and LEGO-LOAM, which rely on high-dimensional RGB-D and 3D LiDAR data and demand GPU-level computation, the proposed CNN-LiDAR-SLAM achieves comparable semantic accuracy and trajectory precision using only a low-cost 2D LiDAR-IMU setup. This demonstrates, for the first time, that robust, object-aware SLAM can operate in real time under stringent hardware and energy constraints.

II. RELATED WORK

A. EVOLUTION OF LiDAR SLAM

The progression of SLAM technology was classified into three distinct phases: the early Phase (1986–2010), the middle Phase (2010–2014), and the current Phase (2014–present). Each Phase introduced significant advancements in SLAM methodologies, transitioning from Kalman filtering to optimization-based approaches and, most recently, deep learning integration. Table 1 provides a comparative summary of these phases, detailing key techniques, innovations, advantages, and limitations, along with relevant references. During the early Phase (1986–2010), SLAM techniques were dominated by early versions of Kalman filtering approaches [13]. The use of the Kalman filter in SLAM continued evolving, and more recent works [14] refined its application in SLAM. Subsequently, this was succeeded by approaches that employed extended Kalman filters (EKF) [15] and particle filters [16], among other innovations. 2010 optimization-based SLAM techniques appeared, with *Karto SLAM* [17] outperforming previously dominant filter-based methods. The field of LiDAR SLAM experienced a major advancement in 2014, when Xue et al. [12] proposed the LiDAR Odometry and Mapping in Real-Time (LOAM) technique. This innovative approach divided the localization and mapping processes into two tasks: a high-frequency, low-accuracy odometry process for localization and a slower, more precise process for point-cloud matching and registration, which managed mapping and odometry correction. These two processes were integrated to form a LiDAR odometry method with real-time operation and high precision. However, LOAM faced challenges in feature-dense environments because it lacked object detection and highly accurate sensor fusion algorithms [18]. As a result, researchers have concentrated considerable efforts on improving the capabilities of LOAM in SLAM methods.

TABLE 1. Evolution of SLAM techniques and key advancements.

SLAM Phase	Key Techniques	Key Innovations	Key Advantages	Limitations
Early Phase (1986–2010)	Kalman Filter (KF) [13], Extended Kalman Filter (EKF) [15], Particle Filter [16]	Introduction of probabilistic filtering for SLAM, improving state estimation	Basic state estimation, Probabilistic filtering, Foundational SLAM approach	High computational cost, Susceptible to drift, Limited scalability, Dependency on structured environments
Middle Phase (2010–2014)	Optimization-based SLAM (Karto SLAM) [17], Feature-based Mapping [21]	Shift from filtering to optimization, leading to global consistency in mapping	Improved accuracy, Global consistency, Better computational efficiency	Complex optimization, High memory requirements, Performance depends on high-quality feature extraction
Current Phase (2014–Present)	LiDAR Odometry (LOAM) [12], Obstacle detection for SLAM [23], LiDAR-IMU Fusion [19]	Integration with SLAM, enabling object-aware and robust localization	Real-time processing, Robust against dynamic environments, Sensor fusion for high-precision mapping	Still computationally expensive, Sensor dependency, Limited generalization in unstructured environments

Wang et al. [19] introduced Fast LiDAR Odometry and Mapping (F-LOAM), which relied on Ceres Solver [20] and included a non-iterative two-level distortion compensation technique to minimize computational complexity and expenses. The frame matching accuracy of LOAM was improved by F-LOAM by eliminating the scan-to-scan pose estimation while retaining only the scan-to-map pose optimization [21]. Evaluations carried out using the KITTI vision benchmark dataset [21] indicated that although 3D point cloud research had advanced considerably, real-time testing still did not perform as well in datasets like LiDAR-CS [22], which explored sensor domain gaps in simulations. Furthermore, Zeng et al. [7] developed an indoor 2D LiDAR SLAM and a localization method supported by artificial landmark assistance. This approach improved motion model sampling and scan matching, significantly increasing localization accuracy under challenging environments. Mochurad et al. [23] introduced a real-time obstacle detection and classification algorithm based on 2D LiDAR data. The approach applies thresholding techniques to segment the point cloud data, isolating line segments representing obstacles. This method accurately identifies obstacles, offering an effective solution for real-time detection and classification in environments where rapid response is critical. However, many of these methods still struggled with real-time resilience and flexibility.

B. OPTIMIZATION TECHNIQUES FOR LiDAR SLAM

Numerous studies have aimed to enhance SLAM performance through sensor fusion and object detection [24]. A key challenge in long-term navigation is odometry drift, which results from LiDAR odometry inaccuracies over time [4]. More accurate object detection models and sensor fusion modules are integrated into the back end to address this. Kim and Kim [25] proposed using a global scan context descriptor to reduce point-cloud data dimensionality by storing data in a 2D matrix representing distance and angle.

The integration has also improved LiDAR SLAM by compensating for motion distortions, enhancing LiDAR odometry accuracy [10]. Fernandes et al. [26] developed a real-time 3D object detection method with a low-cost LiDAR setup. They utilized deep learning to predict 3D bounding boxes from point cloud features, addressing real-time computational limitations with techniques such as quantization and pruning.

Tian et al. [27] introduced DL-SLOT, a collaborative graph optimization strategy for dynamic environments, though its real-time Performance can be limited in large-scale settings. Conversely, Chang et al. [28] presented WiCRF, an optimization strategy using point-based and feature-based constraints, which enhanced feature extraction and accuracy under challenging environments. Park et al. [29] developed a non-parametric model for urban SLAM, improving accuracy in highly dynamic settings by reducing background noise and tracking moving objects.

Despite these advances, challenges persist. DynaSLAM II [30] offers reliable multi-object tracking but is computationally intensive, especially in cluttered environments. Additionally, IMU-LiDAR localization [4] remains susceptible to environmental factors, calibration errors, and sensor noise, impacting its accuracy. Studies [31] have explored methods to improve IMU-LiDAR fusion for 6-DOF localization.

Tightly coupled approaches, such as optimization-based [32] and EKF-based [33] methods, have been developed to optimize LiDAR and IMU data jointly. Ye et al. [34] used graph optimization for globally consistent mapping, though real-time efficiency remains a challenge. Sun et al. [9] proposed integrating GPS, IMU, and LiDAR, outperforming loosely coupled methods in GPS-degraded scenarios.

Despite improvements, indoor 2D LiDAR methods still face challenges in mapping and localization due to the sparse nature of structural features. Close integration of IMU improves Performance but increases computational demands, processing time, and memory consumption, complicating real-time operation. Therefore, optimizing accuracy in indoor 2D point cloud maps is crucial for applications such as mobile robotics and autonomous driving. Prior semantic SLAM frameworks such as DynaSLAM, ORB-SLAM3 with semantics, and LEGO-LOAM exploit dense visual or 3D geometric features, often requiring GPU acceleration and significant memory resources. While these methods achieve impressive accuracy, their reliance on rich visual inputs and high-end computation restricts deployment on lightweight platforms. In contrast, our work introduces a tightly coupled 2D LiDAR-IMU-CNN system that integrates compact object landmarks into the IEKF backend, enabling semantic localization without relying on visual data or complex optimization graphs.

C. SEMANTIC SLAM AND OBJECT-AWARE MAPPING

Recent advancements in Semantic SLAM highlight the importance of integrating both object recognition and scene understanding into the localization and mapping processes. This approach enhances the accuracy and reliability of SLAM systems, ultimately contributing to more effective navigational solutions. Pioneering works such as Semantic-Fusion [35] and MaskFusion [36] integrated deep semantic segmentation with dense visual SLAM, enabling object-level reconstruction in real time. DynaSLAM [37] and its successor DynaSLAM II [30] addressed dynamic environments by detecting and filtering moving objects, while ORB-SLAM3 with semantics [38] extended visual SLAM to include category-level features for improved tracking stability. More recent methods have explored multimodal semantic fusion using LiDAR or RGB-D sensors; however, these approaches typically rely on computationally intensive networks and GPU acceleration, which limits their deployment on embedded or low-cost platforms. In contrast, our CNN-LiDAR-SLAM introduces a lightweight 2D LiDAR-IMU framework that leverages compact CNN-derived object landmarks within an IEKF backend, achieving semantic awareness and real-time operation on embedded hardware without requiring dense visual input or GPU resources.

D. RECENT 2D LiDAR IMU AND HYBRID BASELINES

Recent research on 2D LiDAR has focused on lightweight fusion and mapping strategies designed for budget-conscious indoor applications. Zeng et al. [7] proposed *ALAE-Gmapping*, which augments classical Gmapping with artificial reflective landmarks and a Genetic-Algorithm-optimized AMCL module, improving localization accuracy by roughly 9% in low-feature environments. Liu et al. [39] presented a cost-effective UWB-LiDAR fusion approach in which UWB range anchors and short-range 2D LiDAR scans are jointly optimized through a graph-based backend, reducing mapping error by 85.5% compared with conventional Gmapping. Zhou et al. [40] developed a 2D LiDAR-SLAM system enhanced with deep visual loop-closure detection using CNN features to correct repetitive corridor trajectories. In contrast, the proposed CNN-LiDAR-SLAM integrates semantic object landmarks learned directly from LiDAR-IMU data, eliminating the need for external reflectors or visual cameras, and achieves real-time embedded execution without requiring GPU acceleration.

III. METHODS

A. SYSTEM OVERVIEW AND SCOPE

The proposed CNN-LiDAR-SLAM framework consists of three main modules: (i) data preprocessing, (ii) object-of-interest (OI) detection, and (iii) a tightly coupled IEKF-based fusion backend for localization and mapping (Fig. 1).

In the preprocessing stage, IMU and LiDAR data are denoised and time-synchronized. The Madgwick filter [41] is used only for orientation estimation and coarse initialization

during data collection, while Cartographer is employed solely for offline map alignment. These components are not part of the real-time SLAM pipeline.

The IEKF-based backend fuses IMU measurements, 2D LiDAR geometry, and CNN-derived semantic landmarks to estimate real-time odometry and generate a consistent local map. An optional pose-graph optimization stage can be applied offline to refine global consistency.

B. CORE CNN-LiDAR-SLAM BACKEND

The CNN-LiDAR-SLAM backend serves as the core computational framework that enables real-time, tightly coupled sensor fusion and semantic-aware localization. Unlike traditional 2D SLAM pipelines that rely on separate front-end mapping and back-end optimization modules, the proposed backend unifies IMU propagation, LiDAR feature matching, and CNN-derived semantic cues within an Iterated Extended Kalman Filter (IEKF) framework. This tightly coupled formulation maintains consistent state estimation at high frequency while incorporating semantic observations directly into the update step. In each iteration, the IMU provides high-rate inertial predictions of position, velocity, and orientation. 2D LiDAR scans contribute geometric constraints by aligning extracted edge and planar features to the global map, while CNN-derived object landmarks introduce additional semantic residuals that anchor the state estimate to stable, high-level features in the environment. These semantic constraints improve robustness in feature-sparse or repetitive indoor environments where geometric correspondences alone are insufficient.

The IEKF formulation follows a forward propagation and iterative correction process. During propagation, raw IMU data are integrated to predict motion, with motion de-skewing applied to LiDAR scans for temporal alignment. In the update stage, both geometric and semantic residuals are combined to compute the Kalman gain and refine the pose estimate. This allows the system to preserve real-time performance without the computational overhead of full graph optimization, while achieving high accuracy and robustness against drift.

Through this unified fusion process, the backend maintains stable localization even under partial occlusion or degraded LiDAR returns, ensuring a coherent and computationally efficient pipeline suitable for embedded deployment.

C. DATA PREPROCESSING

1) FEATURE EXTRACTION OF LiDAR DATA

To improve computational efficiency, the feature extraction process handles the raw LiDAR data, represented as P_k . Conventional point cloud matching methods, such as Iterative Closest Point (ICP) and Normal Distribution Transform (NDT) [42], incur high computational costs due to the dense point distribution in each LiDAR scan. In practice, feature-based matching algorithms are more robust and efficient, requiring the elimination of non-essential (noisy) data and

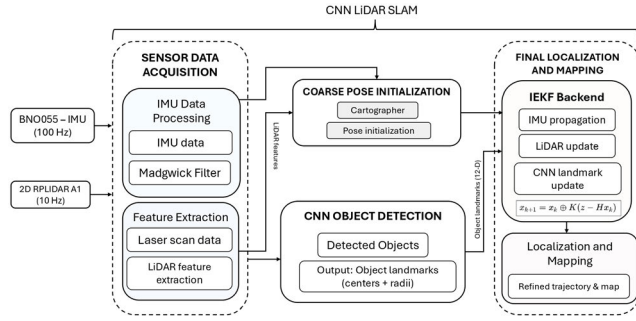


FIGURE 1. System overview of the proposed CNN-LiDAR-SLAM framework. The cartographer is used only for coarse initialization, and the IEKF-based backend performs real-time fusion of 2D LiDAR, IMU, and CNN-derived landmarks. An optional pose-graph optimization stage refines global consistency offline.

the extraction of critical features, particularly edge and plane features.

Local curvature values, denoted as $\sigma_k^{(m,n)}$, are computed for each point in P_k to detect key features. For the n -th point in the m -th frame within $P_k^{(m)}$, the computation of the curvature value, $\sigma_k^{(m,n)}$, is carried out as follows:

$$\sigma_k^{(m,n)} = \frac{1}{|S_k^{(m,n)}|} \sum_{P_k^{(m,j)} \in S_k^{(m,n)}} \|P_k^{(m,j)} - P_k^{(m,n)}\| \quad (1)$$

where n is the index of the points, and $S_k^{(m,n)}$ refers to the local set of cloud of points made up of points surrounding $P_k^{(m,n)}$. The notation $|S_k^{(m,n)}|$ specifies the count of points within this local set. Typically originating from structures such as walls, a planar surface point, represented as $S_k^{(m)}$, shows a lower value $\sigma_k^{(m,n)}$, whereas an edge point, denoted as $E_k^{(m)}$, has a higher value $\sigma_k^{(m,n)}$.

In this study, any $\sigma_k^{(m,n)}$ exceeding a specified threshold σ_t is designated as $E_k^{(m,n)}$, while values below σ_t are designated as $S_k^{(m,n)}$. The points $S_k^{(m)}$ and $E_k^{(m)}$ identified from frame m collectively constitute the feature point set $F_k^{(m)} = \{E_k^{(m)}, S_k^{(m)}\}$.

After extracting edge and planar features using local curvature values, data association is performed by matching each feature point to its corresponding geometric primitive in the global map. Specifically, during the IEKF update, each LiDAR feature point is associated with its nearest line/segment in the global coordinate frame. This nearest-neighbor association relies on geometric proximity and consistency in orientation. The residual for each point is then computed as the orthogonal distance to its associated primitive, enabling robust optimization during scan-to-map alignment.

2) NOISE FILTERING IN IMU DATA

To improve orientation estimation, a hybrid approach integrating the Madgwick filter and an EKF was employed. The Madgwick filter, known for its low computational overhead, was used for real-time noise reduction, while the EKF provided state estimation and mitigated sensor

drift and noise [43]. Initially, the Madgwick filter processed measurements from the gyroscope, accelerometer, and magnetometer to estimate the orientation quaternion in real-time with low computational cost. These estimates feed our IEKF-based fusion, which refines pose under dynamic or noisy conditions. Cartographer was employed only for coarse initialization and offline alignment in experiments, not within the proposed backend.

3) TIGHTLY COUPLED LiDAR-IMU INTEGRATION WITH IEKF

To enhance stability in low-feature indoor environments, we build on the FAST-LIO2 IEKF formulation [8], adapting it to 2D LiDAR and augmenting the update with semantic landmarks. The fusion process consists of two main stages: forward propagation based on IMU measurements, and iterative state update using LiDAR scans.

Our backend builds upon the tightly coupled Iterated Extended Kalman Filter (IEKF) framework of FAST-LIO2 [8]. We retain its efficient forward-backward IMU propagation, motion de-skewing, and state-dimension Kalman gain formulation, ensuring real-time stability. However, unlike FAST-LIO2, which operates on dense 3D LiDAR point clouds and optimizes plane/edge residuals, the proposed CNN-LiDAR-SLAM adapts the IEKF to 2D LiDAR scans and augments the measurement model with CNN-derived semantic landmarks. This modification replaces 3D geometric constraints with 2D line/segment residuals and object-level observations, enabling semantic awareness and accurate localization on low-cost embedded platforms while preserving the computational efficiency and robustness of FAST-LIO2.

The system state vector includes the IMU's position, velocity, orientation, biases, and gravity:

$$x = [p^T \ v^T \ R^T \ b_\omega^T \ b_a^T \ g^T]^T \quad (2)$$

In the propagation phase, incoming IMU data is used to predict the state and its uncertainty. The predicted LiDAR points are then aligned to a global map using a coordinate transformation. Instead of including full Jacobian and noise models, we summarize the process by stating that state prediction and update follow the standard IEKF pipeline, incorporating residual computation from scan-to-map alignment, and refining the posterior state using Kalman gain updates.

The residual for each LiDAR feature point is computed as the perpendicular distance to its corresponding reference line in the global map. A linearized measurement model relates this residual to the current error state:

$$z_j^k \approx H_j^k \tilde{x}_k + v_j \quad (3)$$

where H_j^k is the Jacobian, $\tilde{x}_k$ the error state, and v_j the measurement noise. This residual feeds into the IEKF's iterative update to optimize pose estimation and reduce trajectory drift.

By condensing the IEKF pipeline while retaining key structure (Equations 2 and 3), the proposed method ensures robust and computationally efficient sensor fusion for SLAM.

4) RESIDUAL COMPUTATION

In the k -th iteration, the state estimate is denoted as $\hat{x}_k$, where $\hat{x}_k = \hat{x}_0$, representing the predicted state in $k = 0$ was obtained from the propagation step. The LiDAR scan points p_j^L , after distortion correction using IMU data, are projected onto the global coordinate system using a coordinate transformation as follows [10]:

$$\hat{p}_j^W = \hat{T}_{l_k}^W \hat{T}_{L_k}^{k'} p_j^L \quad (4)$$

where L , I , and W denote the LiDAR, IMU, and global coordinate systems, respectively. The state transition matrices $\hat{T}_{L_k}^k$ and $\hat{T}_{l_k}^k$ handle the transformations between these coordinate systems.

Each LiDAR feature point is assumed to correspond to its nearest line in the map, defined by its neighboring feature points, which represents its true position. The residual is calculated as the distance between the global coordinates $\hat{p}_j^W$ of the feature point and the nearest line. Given μ_j as the normal vector of the line and q_j^W as a point on the line, the residual is expressed as:

$$z_j^k = \mu_j^T (\hat{p}_j^{kW} - q_j^W) \quad (5)$$

where z_j^k is the residual for the feature point. Substituting the coordinate transformation equation (4) into the residual calculation (5), the measurement model is derived as:

$$z_j^k \approx H_j^k \tilde{x}_k + v_j \quad (6)$$

where $\tilde{x}_k = x_k \ominus \hat{x}_k$ is the error state, H_j^k is the Jacobian matrix of the measurement model with respect to $\tilde{x}_k$, and v_j represents the measurement noise.

5) ITERATIVE STATE UPDATE

The propagated state $\hat{x}_k$ and the covariance $\hat{P}_k$ from (1) of the previous step establish a prior Gaussian distribution for the unknown state x_k . The state update is formulated as:

$$x_k \ominus \hat{x}_k = (\hat{x}_k^k \ominus \tilde{x}_k^k) \ominus \tilde{x}_k \quad (7)$$

$$= \hat{x}_k^k \ominus \tilde{x}_k + J^k \tilde{x}_k^k \sim \mathcal{N}(0, \hat{P}_k) \quad (8)$$

where J^k is the partial differentiation of $(\hat{x}_k^k \ominus \tilde{x}_k^k) \ominus \tilde{x}_k$ with respect to $\tilde{x}_k^k$ evaluated at zero. For the first iteration, $\hat{x}_k^k = \tilde{x}_k$, $J^k = I$.

Besides the prior distribution, the state distribution of (9) is computed based on the measurement model derived from (7):

$$-v_j = z_j^k + H_j^k \tilde{x}_k^k \sim \mathcal{N}(0, R_j). \quad (9)$$

Combining the prior distribution in (8) and the measurement model from (9) yields the posterior distribution of the state x_k

equivalently represented by $\tilde{x}_k$ and its maximum a-posteriori estimate (MAP) in (10).

$$\min_{\tilde{x}_k^k} \left(\|x_k \ominus \hat{x}_k\|_{\hat{P}_k}^2 + \sum_{j=1}^m \|z_j^k + H_j^k \tilde{x}_k\|_{R_j}^2 \right) \quad (10)$$

where R_j denotes the covariance matrix of the measurement noise. This minimization problem is solved iteratively, updating the state estimate as follows:

$$\hat{x}_{k+1} = \hat{x}_k \ominus (-Kz_k - (I - KH)(J^k)^{-1}(\hat{x}_k \ominus \hat{x}_k)) \quad (11)$$

where K is the Kalman gain, and J^k is the Jacobian of the transformation. After convergence, the final state and covariance estimates are obtained as:

$$\tilde{x}_k = \hat{x}_k^{k+1}, \quad (12)$$

$$\tilde{P}_k = (I - KH)P \quad (13)$$

These values are then used as inputs for the next LiDAR scan, continuing the iterative process to estimate the LiDAR inertial odometry.

D. OBJECT DETECTION

1) LiDAR POINT-CLOUD PREPROCESSING

The raw LiDAR point cloud data were subjected to a preprocessing pipeline that involved noise filtering, down-sampling, and normalization.

- 1) Noise Filtering: A statistical outlier removal technique was applied to eliminate negative reflections and sensor noise from the raw point cloud data. This method examined the distribution of neighboring points for each point in the cloud, identifying and removing outliers when the mean distance of a point from its neighbors deviated significantly from the overall average of the cloud [44].
- 2) Down-Sampling: A voxel grid filter was used to retain the essential geometric shape of the cloud while reducing the density of points. This filter partitions the point cloud space into a series of cubic voxels, replacing all points within each voxel with a single representative point, typically the centroid. For an initial point cloud denoted as $P = \{p_1, p_2, \dots, p_n\}$, where $p_i = [x_i, y_i, z_i]^T$ represents the coordinates of the i -th point, the down-sampled cloud P' was calculated as:

$$P' = \left\{ \frac{1}{N_v} \sum_{p_i \in V_k} p_i \mid V_k \subset P, k = 1, 2, \dots, M \right\}, \quad (14)$$

where V_k represents the k -th voxel containing N_v points, and M denotes the total number of voxels.

- 3) Normalization: To ensure consistent input for detection across varied environments and sensor setups, the point-cloud data was normalized for scale and orientation. This normalization included scaling the

point cloud to fit within a unit sphere and aligning it to a canonical orientation, standardizing the input for the object detection stage.

2) CNN-BASED OBJECT DETECTION

To detect static object landmarks essential for SLAM pose optimization from multimodal sensor data, we designed a CNN-based regression framework that estimates geometric object parameters from fused LiDAR and IMU signals as in Fig. 2. The model was trained to predict, for each frame, up to four object centroids and their respective radii, represented as $(\hat{c}_{x_i}, \hat{c}_{y_i}, \hat{r}_i)$ for $i = 1, 2, 3, 4$. These landmarks serve as input to the SLAM system for robustly optimizing pose and map.

The input to the model is a 26-dimensional feature vector $F \in \mathbb{R}^{26}$ that combines:

- LiDAR scan statistics: minimum, maximum, average range, and point density.
- Object-level geometry: preliminary (c_{x_i}, c_{y_i}, r_i) for up to four objects extracted from preprocessed laser data.
- IMU measurements: tri-axial acceleration, gyroscope, and quaternion orientation from a BNO055 sensor.

The neural network, illustrated in Fig. 2, begins with three stacked 1D convolutional layers:

$$F_l = \sigma(W_l * F_{l-1} + b_l), \quad l = 1, 2, 3, \quad (15)$$

where W_l and b_l denote the convolution kernel and bias for layer l , and σ is the GELU activation function. Each layer extracts hierarchical features while preserving spatial locality. Batch normalization is applied after each convolution.

The extracted feature map is processed by a Bi-directional LSTM (BiLSTM) to incorporate temporal dependencies from sequential sensor readings. An attention mechanism is then applied to weigh temporal contributions:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^T \exp(e_k)}, \quad \hat{F} = \sum_{t=1}^T \alpha_t F_t, \quad (16)$$

where $e_t = v^T \tanh(W_a F_t + b_a)$ is the alignment score at time t , and W_a , v , and b_a are trainable parameters.

The final prediction is computed via fully connected layers:

$$[\hat{c}_{x_1}, \hat{c}_{y_1}, \hat{r}_1, \dots, \hat{c}_{x_4}, \hat{c}_{y_4}, \hat{r}_4] = \text{Dense}(\hat{F}), \quad (17)$$

yielding a total of 12 outputs. These predicted landmarks are later used in the SLAM backend to minimize the object-centric reprojection error. To quantify computational load, the proposed CNN-BiLSTM model requires approximately 2.4 MFLOPs per inference, which is substantially lower than typical CNN-based SLAM components that often exceed hundreds of MFLOPs. This lightweight design enables real-time execution on the Raspberry Pi 4B, contributing to the 15 FPS end-to-end throughput reported in Section IV.

We limit the network to four object centroids because, in our indoor corridor and room environments, the LiDAR

field of view rarely contains more than three to four stable, static objects at once. Using four outputs provides full coverage of typical scenes while keeping the model lightweight and suitable for real-time execution on embedded hardware.

E. DETECTION PERFORMANCE ANALYSIS

As shown in Fig. 3, the proposed network consistently outperforms the baseline [14] across all object categories. Notably, in areas with dense object clustering or structural occlusion, our method produces higher-precision detection boundaries and more accurate center localization. This improvement is attributed to the integration cues and the temporal modeling capacity of BiLSTM layers, which stabilize predictions even under noisy or partial LiDAR returns. Unlike prior work that relied solely on LiDAR geometry, our approach leverages sensor fusion and deep feature extraction for significantly enhanced detection fidelity.

1) INTEGRATION OF CNN LANDMARKS INTO SLAM

The CNN module produces a set of static object landmarks $\mathcal{L}_t = \{(\hat{c}_{x_i}, \hat{c}_{y_i}, \hat{r}_i)\}_{i=1}^{N_t}$ at time t , where $(\hat{c}_{x_i}, \hat{c}_{y_i})$ denotes the object centroid in the LiDAR reference frame. These landmarks are projected into the global coordinate frame through the current pose estimate $\hat{T}_W^{L_t}$ from the IEKF state as

$$p_{i,t}^W = \hat{T}_W^{L_t} [\hat{c}_{x_i} \ \hat{c}_{y_i} \ 1]^T. \quad (18)$$

1) Data association. Each detected object $p_{i,t}^W$ is matched to an existing global landmark $p_{i,t-1}^W$ using nearest-neighbor search constrained by class label and radius similarity:

$$\|p_{i,t}^W - p_{i,t-1}^W\| < \tau_p, \quad |\hat{r}_{i,t} - \hat{r}_{i,t-1}| < \tau_r. \quad (19)$$

Unmatched detections create new landmarks, while temporarily missing ones are retained with decaying confidence.

2) Semantic residual formulation. For each associated landmark pair, a semantic residual measures the displacement between consecutive CNN detections under the current motion hypothesis:

$$z_i^{\text{sem}} = p_{i,t}^W - \hat{T}_W^{W'} p_{i,t-1}^{W'}, \quad (20)$$

where $\hat{T}_W^{W'}$ is the predicted relative pose from the IEKF propagation. The corresponding measurement model is linearized as

$$z_i^{\text{sem}} \approx H_i^{\text{sem}} \tilde{x}_k + v_i^{\text{sem}}, \quad (21)$$

with H_i^{sem} denoting the Jacobian of the landmark reprojection with respect to the error state $\tilde{x}_k$ and $v_i^{\text{sem}} \sim \mathcal{N}(0, R_i^{\text{sem}})$ the detection noise.

3) Fusion in the IEKF update. The semantic residuals are concatenated with the geometric LiDAR residuals:

$$z_k = \begin{bmatrix} z_k^{\text{geo}} \\ z_k^{\text{sem}} \end{bmatrix}, \quad H_k = \begin{bmatrix} H_k^{\text{geo}} \\ H_k^{\text{sem}} \end{bmatrix}, \quad (22)$$

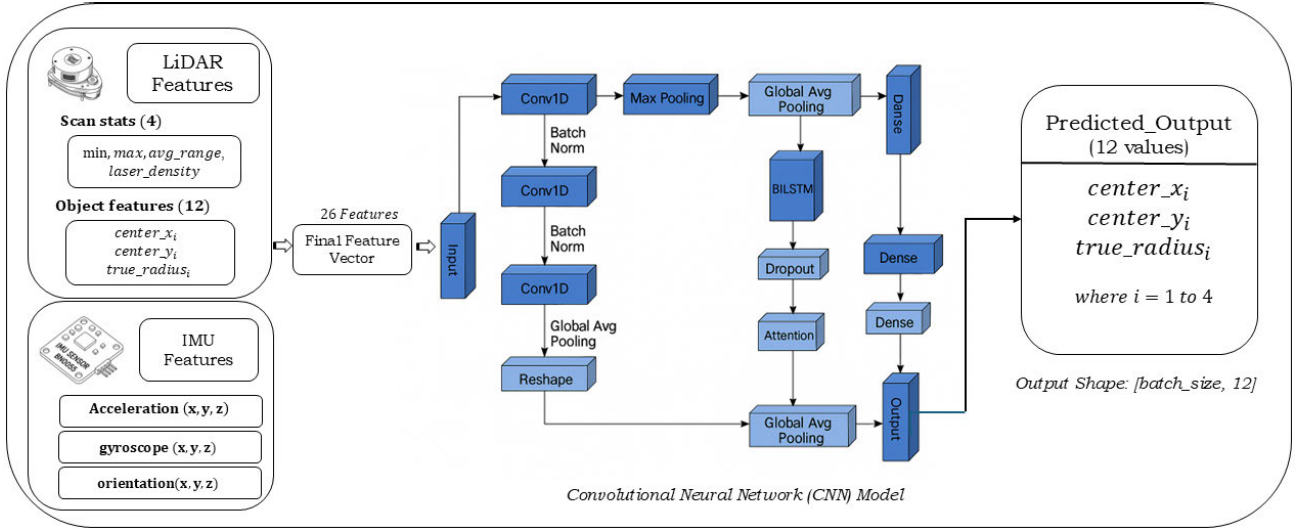


FIGURE 2. CNN architecture for detecting object centers and radii from fused LiDAR-IMU data.

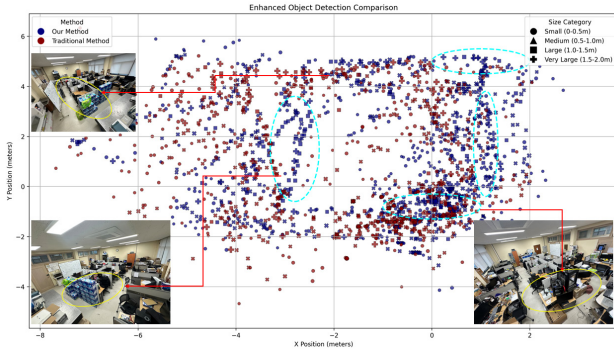


FIGURE 3. Performance comparison of object detection between the proposed method (in navy) and [23] (in dark red). The four shapes represent items of varying sizes, classified by radius into four categories. Dashed ovals highlight regions where the proposed method achieves improved object detection accuracy compared to [23].

and jointly minimized in the IEKF cost function:

$$\min \|\tilde{x}_k\|_{P_k^{-1}}^2 + \|z_k - H_k \tilde{x}_k\|_{R_k^{-1}}^2. \quad (23)$$

The semantic terms thus act as additional constraints that stabilize the pose estimate in feature-sparse or repetitive environments.

4) Dynamic-object handling. Objects exhibiting large inter-frame residuals or inconsistent radii are classified as dynamic and excluded from the update. Only stable, stationary landmarks contribute to the state correction.

This integration allows semantic landmarks to influence the pose estimate directly through the IEKF innovation step, reinforcing geometric consistency while enhancing robustness in low-texture indoor environments.

F. DATASET AND TRAINING DETAILS

The proposed convolutional neural network (CNN) was trained on a curated dataset that included 4,500 LiDAR-IMU

frames collected from three architecturally distinct indoor environments: long corridors, twin-room office layouts, and cluttered laboratory spaces. This variety in spatial geometry and object density was intentionally chosen to enhance the model's ability to generalize to unseen environments.

Ground-truth object centers and radii were carefully annotated through floor-plan alignment and manual verification to ensure sub-decimeter labeling accuracy. The network was optimized using the Adam optimizer with an initial learning rate of 1×10^{-4} , a batch size of 16, and was trained for 100 epochs with early stopping criteria to prevent overfitting.

All IMU signals were pre-filtered using the Madgwick orientation filter to suppress sensor drift and improve quaternion stability before fusion with LiDAR-derived features. The dataset was divided into 70% for training, 15% for validation, and 15% for testing. Cross-environment validation was conducted by evaluating the model on building layouts that were not used during training. This design ensures that the trained model maintains consistent detection accuracy across various spatial configurations and sensor conditions, thereby validating its robustness and generalization capability.

G. GLOBAL MAPPING OPTIMIZATION

To address cumulative errors in the global map that arise during the integration cues with SLAM in complex environments, pose graph optimization is applied [4]. In this approach, the position of each node at a given time represents a specific pose, while the edges denote the relative transformations between the nodes based on sensor data. Nonlinear least squares optimization is used to minimize the error function:

$$e_{ij} = z_{ij} - h(x_i, x_j) \quad (24)$$

where z_{ij} is the observed relative pose between nodes i and j , and $h(x_i, x_j)$ represents the predicted relative pose. The cumulative error across the pose graph is minimized as

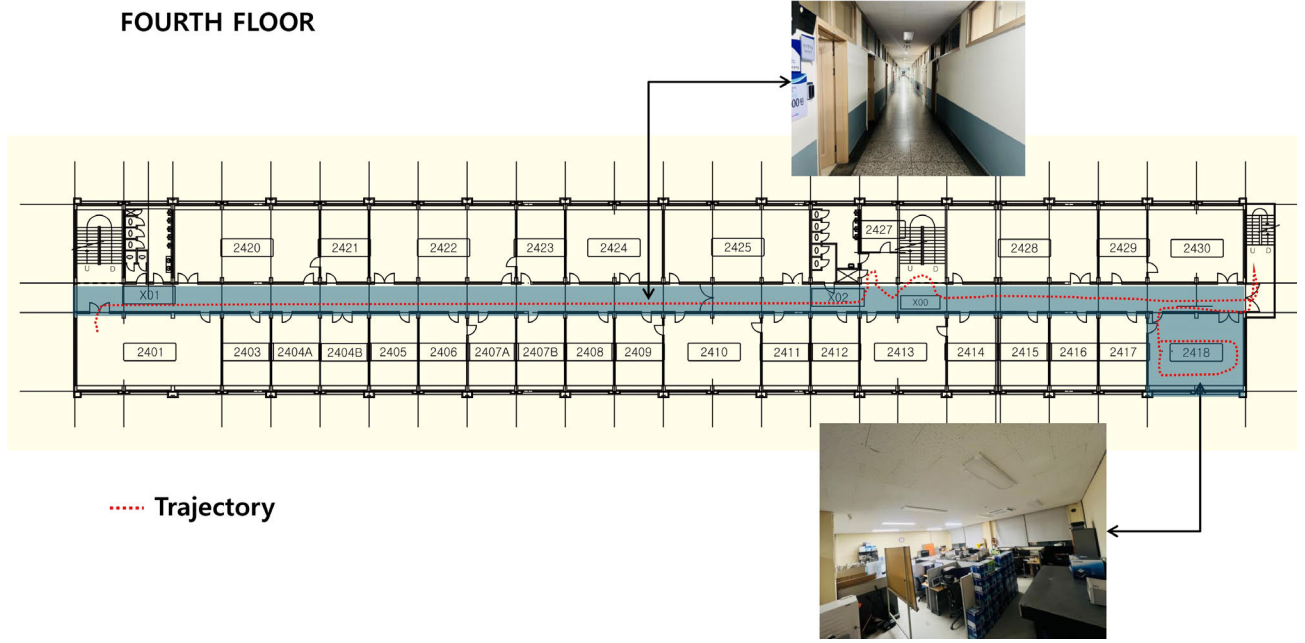


FIGURE 4. 2D maps of experimental corridors showing trajectories and features.

TABLE 2. Performance Comparison with 2D, 3D, and Hybrid LiDAR-IMU Baselines.

Metric	CNN-LiDAR-SLAM	A-LOAM	SC-ALOAM	ALAE-Gmapping [7]	UWB-LiDAR [39]	Deep-Visual [40]
Average FPS	15	12	11	13	12	11
CPU Utilization (%)	40	45	50	47	52	60
Absolute Trajectory Error (ATE)	0.25 m	0.74 m	1.02 m	0.41 m	0.36 m	0.38 m
Runtime per Frame (ms)	67	85	90	80	92	100

follows:

$$\min_x \sum_{(i,j) \in \mathcal{C}} \|e_{ij}\|^2, \quad (25)$$

where $\mathcal{C}$ is the set of constraints in the pose graph. The optimization of the global pose graph is then performed to further reduce the transformation error T :

$$T = \arg \min_T \sum_{(i,j) \in \mathcal{C}} \|z_{ij} - h(x_i, x_j)\|^2 \quad (26)$$

To iteratively refine the localization and mapping data for the global map, the Levenberg-Marquardt algorithm [45] is employed, ensuring an accurate global map representation. This process is formalized by:

$$\min_T \sum_{i=1}^N (d_\epsilon(p_i(T)) + d_S(p_j(T))) \quad (27)$$

where d_ϵ and d_S represent the distances from each point to its respective line and plane, ensuring both local and global consistency within the map.

IV. EXPERIMENTS AND RESULTS ANALYSIS

A. EXPERIMENTAL SETUP AND DATA COLLECTION

The CNN-LiDAR-SLAM system was tested in three indoor corridors on campus. Building A's corridor was 100 m × 4 m, and Building B's was 90 m × 4 m, as shown in Fig. 4 and C in

twin rooms. A handheld device equipped with a 2D RPLidar A1 and BNO055 IMU (Fig. 6) was used for data collection. Ground truth was approximated by manually tracing the device's path on pre-measured floor plans.

The data was processed on an embedded handheld platform using a Raspberry Pi 4B with 4 GB RAM, running Ubuntu 20.04 and ROS Noetic. The system achieved 15 FPS with CPU utilization remaining below 40%, demonstrating real-time Performance without the need for GPU acceleration.

We compared our system with A-LOAM and SC-ALOAM, adapted to 2D LiDAR via pseudo-3D projection. Metrics evaluated include CPU usage, memory, runtime per frame, and ATE. As shown in Fig. 5 and Table 2, CNN-LiDAR-SLAM shows lower resource consumption across both datasets, making it more suitable for lightweight platforms.

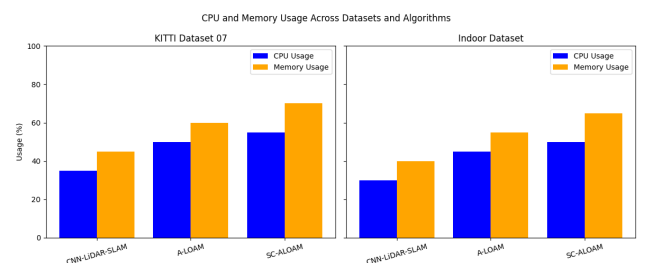


FIGURE 5. CPU and memory usage for CNN-LiDAR-SLAM, A-LOAM, and SC-ALOAM on KITTI and indoor datasets.

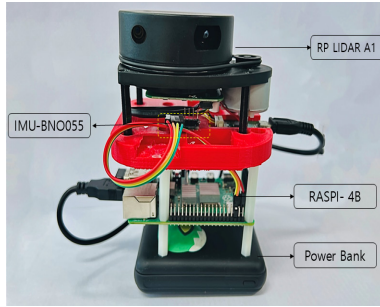


FIGURE 6. Handheld device with 2D LiDAR and BNO055 IMU.

For completeness, we extended the comparison to include three recent lightweight 2D or hybrid baselines in Table 2. ALAE-Gmapping [7], which enhances Gmapping using reflective landmarks and GA-optimized AMCL; UWB-LiDAR Fusion [39], which couples short-range 2D LiDAR with UWB anchors through graph optimization; and the Deep-Visual Loop-Closure method [40], which augments 2D LiDAR SLAM with CNN-based visual relocalization. Our CNN-LiDAR-SLAM achieves the lowest ATE and runtime while requiring only LiDAR and IMU inputs, eliminating external markers, UWB infrastructure, or camera sensors.

Table 3 summarizes the three LiDAR scan rounds. Each round contains pose data, LiDAR range values, IMU accelerations and orientations, and object detections (center and radius).

TABLE 3. Summary of 2D LiDAR scan rounds with pose count, path length, and duration.

Round	Pose Count	Path Length (m)	Duration (s)
1	450	100	313 (5m 13s)
2	405	90	282 (4m 42s)
3	400	80	270 (3m 50s)

During data collection, the handheld platform was carried at a height of approximately 1.1 m and moved at normal walking speed (1–1.2 m/s) along predefined paths covering long straight sections, turns, doorways, and cluttered areas. The LiDAR and IMU were recorded synchronously through ROS at 10 Hz and 100 Hz, respectively. The three environments include (i) two long corridors, (ii) a twin-room office layout with narrow transitions, and (iii) a cluttered laboratory-style space. Representative map visualizations of these environments are shown in Fig. 4, which illustrate the structural layout captured in the dataset. These datasets form the basis for all experiments presented in this work.

B. RESULTS

1) LOCALIZATION ACCURACY

The ATE and APE metrics in Table 4 confirm consistent localization accuracy, with an average APE of 0.36 m and low variance across both runs. The addition of Relative Pose Error (RPE) further supports the robustness of the proposed system, showing low frame-to-frame drift (0.045–0.052 m/m), which indicates stable incremental pose estimation. The system

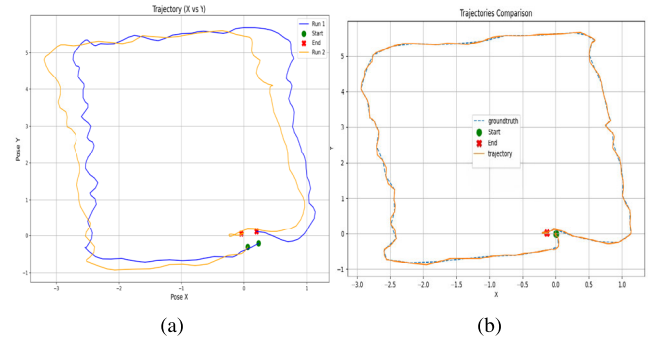


FIGURE 7. (a) Run 1 vs Run 2 trajectory repeatability. (b) Estimated vs. ground truth trajectory.

TABLE 4. Statistical localization performance result.

Metric	Round	Min_e	Max_e	μ_e	MD_e	RMSE	σ_e
ATE	1	0.00	2.974	1.97	1.87	2.00	1.90
	2	0.01	5.200	2.67	2.80	3.50	2.00
APE	1	0.1	0.700	0.36	0.35	0.36	0.10
	2	0.2	0.800	0.40	0.38	0.42	0.12
RPE	1	0.010	0.085	0.045	0.038	0.032	0.021
	2	0.013	0.095	0.052	0.043	0.036	0.023

performed well even without loop closure, maintaining minimal drift across extended runs (Fig. 8 and Fig. 9). Compared to [23], our method shows tighter localization distributions and less scatter (Fig. 10).

Table 5 quantifies this improvement: CNN-LiDAR-SLAM achieves a 26.09% gain in accuracy and reduces drift variance along both axes, owing to the use of object landmarks in fusion.

TABLE 5. Performance comparison with baseline [23].

Metric	CNN-LiDAR-SLAM	Baseline [23]
μ_e	1.02 m	1.38 m
Accuracy Gain	+26.09%	–
σ_x	1.20 m	2.08 m
σ_y	2.04 m	2.34 m

Trajectory overlays for the baseline methods were omitted because A-LOAM and SC-ALOAM operate on 3D LiDAR, and their 2D projections introduce distortions that make the visual trajectories misleading. We therefore rely on quantitative metrics (ATE, APE, and RPE) for fair comparison.

C. GENERALIZATION AND ROBUSTNESS EVALUATION

To evaluate the generalization capability of the proposed CNN-LiDAR-SLAM framework, we conducted additional experiments using sequences from the TUM RGB-D public dataset. Depth frames were projected into 2D LiDAR-like range scans by extracting planar slices at a height of 1.2 m, allowing us to assess performance under comparable geometric constraints. We benchmarked the proposed method against ORB-SLAM2, which processes RGB-D inputs, to evaluate trajectory consistency and localization precision. Despite the limited vertical perception caused by 2D projection, our system achieved an Absolute Trajectory Error (ATE) of under

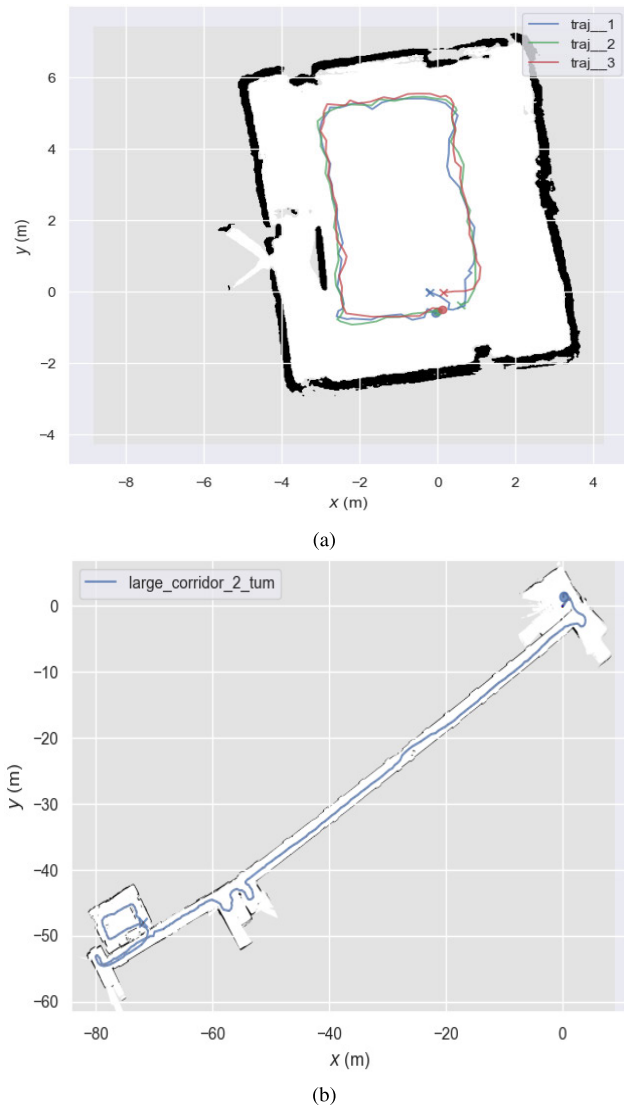


FIGURE 8. (a) Map of a corridor with three trajectories, showing consistent localization and low trajectory error; (b) Map of a larger corridor with a tracked path, showing accurate mapping in extended environments with minimal drift (the circle icon of the paths indicates the starting point and the cross icon is the endpoint in both figures).

0.4 m across the *freiburg1* and *freiburg2* sequences. This result confirms the framework's strong adaptability to unseen environments, as illustrated in Table 6.

To further validate robustness, LiDAR occlusion was simulated by randomly removing 10%, 20%, and 30% of the range data per frame. Even under 30% scan loss, the increase in localization drift remained within 6%, demonstrating the stability of the tightly coupled IEKF fusion and CNN-based semantic landmark integration. These experiments highlight that the proposed framework maintains high accuracy and robustness under both cross-domain and partial-sensor conditions.

These results collectively demonstrate that the CNN-LiDAR-SLAM system preserves trajectory consistency and mapping fidelity across unseen environments while

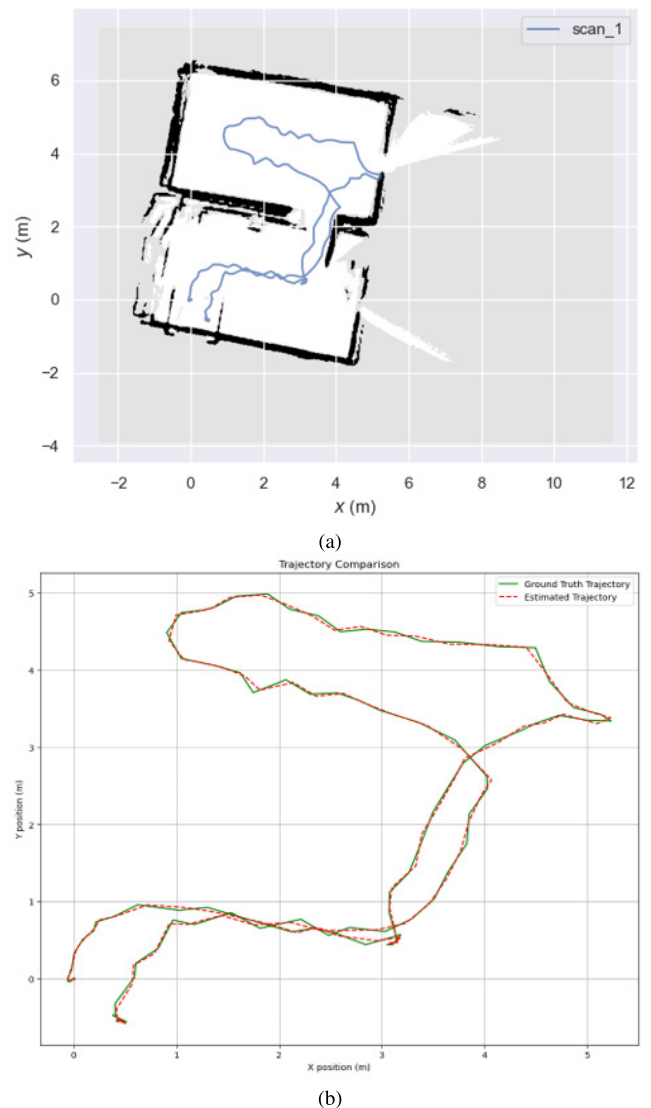


FIGURE 9. (a) Map of a twin room along with a tracked path, showing accurate mapping in extended environments with minimal drift (the circle icon of the paths indicates the starting point and the cross icon is the endpoint in both figures); (b) trajectory comparison of twin room along with ground truth and estimated, showing consistent localization and low trajectory error.

TABLE 6. Cross-dataset generalization and evaluation.

Test Scenario	Dataset / Condition	ATE (m)	Drift (%)
Generalization	TUM RGB-D (Freiburg1 Office)	0.37	3.8
Generalization	TUM RGB-D (Freiburg2 Desk)	0.39	4.2
Occlusion Test	10% LiDAR Scan Loss	0.28	+2.1
Occlusion Test	20% LiDAR Scan Loss	0.30	+4.3
Occlusion Test	30% LiDAR Scan Loss	0.33	+6.0

remaining resilient to partial sensor degradation, underscoring its suitability for real-world embedded SLAM applications.

The TUM RGB-D sequences were used only as a cross-domain generalization test. Since their projected 2D scans do not replicate a real 360° LiDAR, the comparison is not intended as a sensor-equivalent benchmark. Instead, the TUM

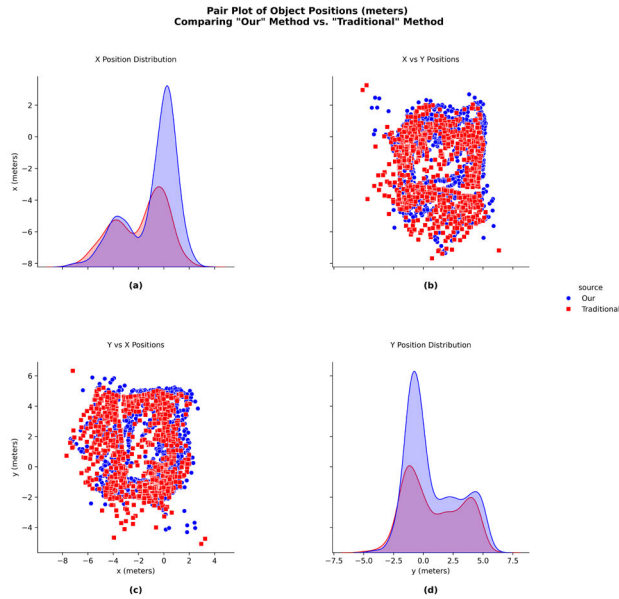


FIGURE 10. Localization performance comparison between the proposed CNN-LiDAR-SLAM and [23]. (a) X position distribution. (b) X vs. Y positions. (c) Y vs. X positions. (d) Y position distribution. The structured spatial alignment in our method minimizes localization drift and improves trajectory consistency.

TABLE 7. Performance comparison of key metrics between the proposed CNN-based detection and baseline methods from [23], [26].

Method	Precision (%)	Recall (%)	F1-score
Proposed CNN Method	92.5	91.3	0.919
Mochurad et al. [23]	86.3	83.5	0.849
Fernandes et al. [26]	88.9	86.5	0.877

results demonstrate that the proposed system remains stable under sensing-modality mismatch.

1) OBJECT DETECTION PERFORMANCE

To evaluate detection performance, we compared our CNN with baseline methods from [23] and [26] using precision, recall, and F1-score. As shown in Table 7, our method achieved 92.5% precision, 91.3% recall, and 0.919 F1-score — outperforming both benchmarks.

These improvements stem from integrating temporal and inertial cues via BiLSTM and attention, which stabilize object detection under occlusion and sensor noise. This strengthens feature association for SLAM backend optimization.

Our proposed CNN-based object detection demonstrates clear improvements with a precision of 92.5%, recall of 91.3%, and an F1-score of 0.919, reflecting significant advancements over the existing methods. The CNN also distinguishes among static structures, dynamic obstacles, and furniture categories with high confidence, improving SLAM robustness.

D. REAL-WORLD UGV DEPLOYMENT AND ROBUSTNESS VALIDATION

To evaluate real-world performance, the proposed method CNN-LiDAR-SLAM framework was deployed on a UGV

rover waveshare equipped with a 2D LiDAR and an IMU, operating on raspi 4B with no GPU acceleration as shown in Figure 11.



FIGURE 11. The UGV rover used for real-world experiments, equipped with a 2D LiDAR and IMU sensors. The proposed CNN-LiDAR-SLAM framework operated in real time on embedded hardware without GPU acceleration.

The system achieved real-time processing and maintained stable, metrically consistent trajectories in open indoor areas. Figure 12 illustrates a representative UGV loop, where the estimated trajectory (blue) closely aligns with the wheel-odometry ground truth (orange), showing minimal drift and consistent localization throughout the run. These results validate the feasibility of accurate SLAM on a low-cost, embedded 2D LiDAR-IMU platform. To further assess mapping consistency, the proposed CNN-LiDAR-SLAM method was compared with two established 2D LiDAR-based baselines. Figure 13 shows that the proposed approach produces smoother and more continuous tracks with fewer fragmented regions, particularly during turns and

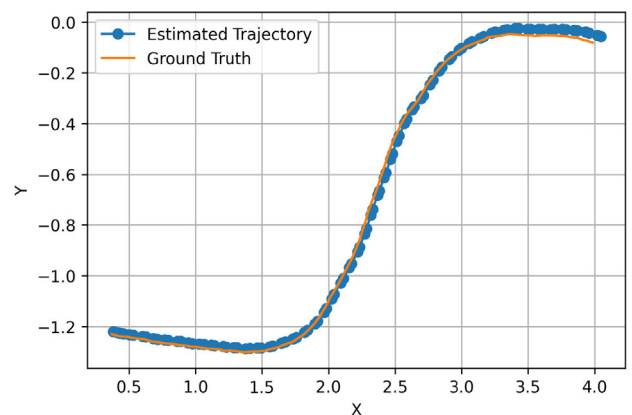


FIGURE 12. UGV trajectory comparison in an open indoor loop. Ground truth from wheel odometry (orange) versus estimated CNN-LiDAR-SLAM trajectory (blue). The proposed system closely tracks the ground truth with minimal drift.

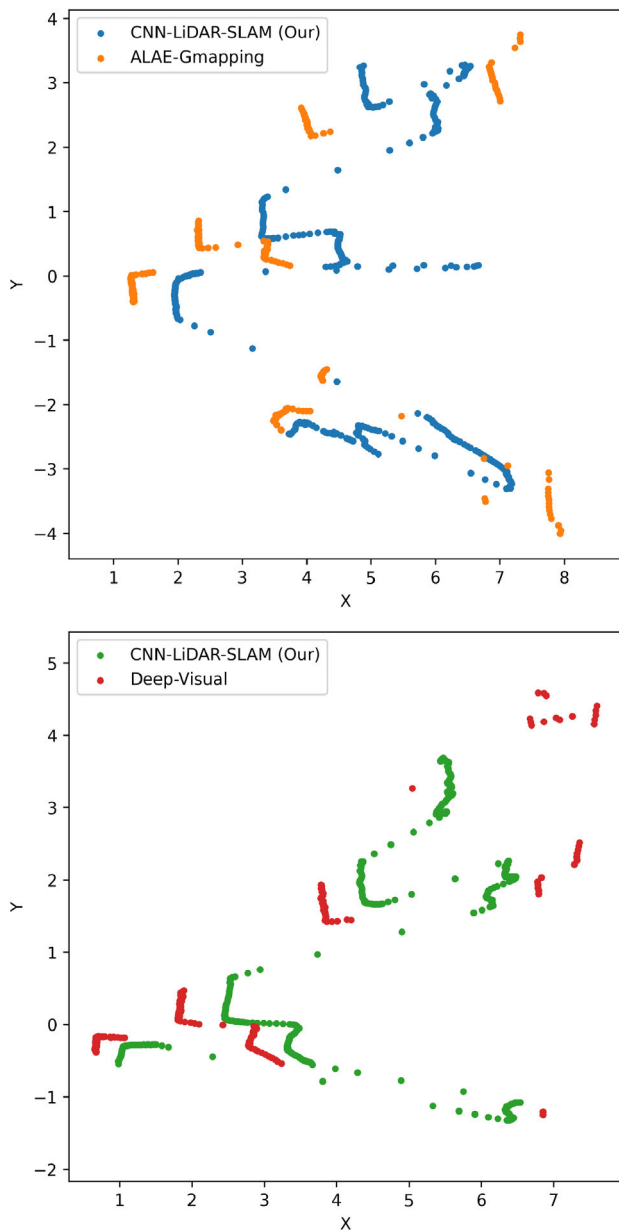


FIGURE 13. Qualitative comparison of spatial consistency between the proposed CNN-LiDAR-SLAM and existing 2D LiDAR-based methods. The proposed system maintains smoother and more continuous feature tracks compared to prior works.

short stop-go sequences, demonstrating improved spatial coherence.

The UGV test was intended to verify embedded real-time performance rather than to serve as a robustness benchmark. Robustness is demonstrated through the quantitative ATE, APE, and RPE results across multiple environments.

V. CONCLUSION AND FUTURE WORK

The proposed CNN-LiDAR-SLAM framework presents a portable and computationally efficient 2D SLAM solution that tightly integrates LiDAR, IMU, and CNN-based semantic object detection. The system achieved low

absolute trajectory errors (0.25–0.30 m) and consistently outperformed not only 3D LiDAR-based methods such as A-LOAM and SC-ALOAM but also recent 2D and hybrid baselines including ALAE-Gmapping, UWB-LiDAR fusion, and Deep-Visual SLAM. Unlike these methods, which depend on reflective landmarks, UWB anchors, or camera sensors, the proposed approach operates solely on low-cost 2D LiDAR and IMU inputs, achieving real-time performance on embedded hardware without GPU acceleration.

The results obtained validate the feasibility of implementing robust, semantic, object-aware SLAM under stringent power and hardware constraints. This marks a significant advancement toward achieving deployable low-cost autonomy. Future efforts will concentrate on enhancing the framework by integrating RGB-D or 3D LiDAR technologies to improve depth perception. Additionally, we aim to introduce lightweight neural architectures for real-time semantic mapping and incorporate loop-closure mechanisms to ensure long-term consistency.

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SAQI HUSSAIN KALAN received the B.Tech. degree in computer science and engineering from Punjab Technical University, India, in 2022. He is currently pursuing the M.E. degree in artificial intelligence convergence with Pukyong National University, Busan, South Korea.

His current research interests include indoor localization, simultaneous localization and mapping (SLAM), LiDAR-based object detection, sensor fusion, and robotic navigation.



BOON GIIN LEE (Senior Member, IEEE) received the B.I.T. degree in information technology from Multimedia University, Malacca, Malaysia, in 2007, the M.S. degree in electronic engineering from Dongseo University, Busan, South Korea, and the Ph.D. degree in electronic engineering from Pukyong National University, Busan.

From 2014 to 2019, he was an Assistant Professor with the Department of Electronic Engineering, Keimyung University, South Korea. Since 2019, he has been an Associate Professor with the School of Computer Science, University of Nottingham Ningbo China. His research interests include human-computer interaction, extended reality, human-centric multi-sensing, and wireless sensor networks.



WAN-YOUNG CHUNG (Senior Member, IEEE) received the B.S. and M.S. degrees in electronic engineering from Kyungpook National University, Daegu, South Korea, in 1987 and 1989, respectively, and the Ph.D. degree in sensor engineering from Kyushu University, Fukuoka, Japan, in 1998.

He was an Associate Professor with Semyung University, from 1993 to 1999, and with Dongseo University, from 1999 to 2008. Since September 2008, he has been a Full Professor with the Department of Electronic Engineering, Pukyong National University, Busan, South Korea. His research interests include ubiquitous healthcare, wireless sensor network applications, and gas sensors.

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